AMATH 353 HW 6

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Question 1: Method of Characteristics (part 1)

(a) Find the characteristic curves of the PDE

$$u_t + xu_x = 0.$$

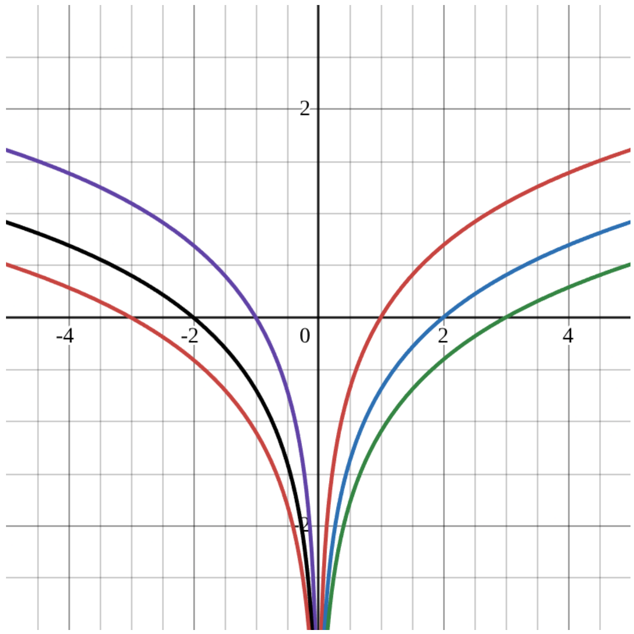
Sketch/plot the characteristic curves in the xt-plane. What do you predict the dynamics tolook like?

$$u_t + xu_x = 0$$
$$\frac{d}{dt}u(x(t), t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + \frac{dx}{dt}u_x = u_t + xu_x$$
$$\frac{dx}{dt} = x$$

Solve the ODE:

$$\frac{1}{x}dx = dt$$
$$\int \frac{1}{x}dx = \int dt$$
$$\ln|x| = t + \ln|x_0|$$
$$x = e^{t+\ln|x_0|}$$
$$x = x_0e^t$$

The wave will go faster the further it gets from the origin. If $x_0 > 0$ it moves to the right. If $x_0 < 0$ it moves to the left.



(b) Use the method of characteristics to solve

$$\begin{cases} u_t + tu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \cos(x), & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$u_t + tu_x = 0$$
$$\frac{d}{dt}u((x),t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + tu_x = u_t + \frac{dx}{dt}u_x$$
$$\frac{dx}{dt} = t$$

Solve the ODE

$$\frac{dx}{dt} = t$$
$$dx = tdt$$
$$\int dx = \int tdt$$
$$x = \frac{t^2}{2} + x_0$$
$$x_0 = x - \frac{t^2}{2}$$
$$u(x_0, 0) = \cos(x - \frac{t^2}{2})$$

$$u(x,t) = \cos(x - \frac{t^2}{2})$$

The wave speeds up as it moves away from x_0, and always moves to the right.

(c) Use the method of characteristics to solve

$$\begin{cases} u_t + \cos(t)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = e^{-x^2}, & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$u_t = \cos(t)u_x = 0$$
$$\frac{d}{dt}u(x(t),t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + \frac{dx}{dt}u_x = u_t + \cos(t)u_x$$
$$\frac{dx}{dt} = \cos(t)$$

Solve the ODE

$$\frac{dx}{dt} = \cos(t)$$
$$dx = \cos(t)dt$$
$$\int dx = \int \cos(t)dt$$
$$x = \sin(t) + x_0$$
$$x_0 = x - \sin(t)$$

$$u(x_0, 0) = e^{-(x-\sin(t))^2}$$
$$u(x,t) = e^{-(x-\sin(t))^2}$$

The wave bounces between -1 and 1, and the velocity depends on time.

Question 2: Method of Characteristics (part 2)

(a) Determine the (implicit) solution to the nonlinear, homogeneous PDE problem

$$\begin{cases} u_t + u^2 u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \tanh(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{aligned} x &= x(t) \\ \frac{d}{dt}u(x(t),t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t = u^2 u_x \\ \frac{dx}{dt} &= u^2 \end{aligned}$$

Solve The ODE

$$\begin{aligned} \frac{dx}{dt} &= u^2 \\ dx &= u^2 dt \\ \int dx &= \int u^2 dt \\ x &= u^2 t + x_0 \\ x_0 &= x - u^2 t \\ u(x_0, 0) &= \tanh(x_0) \\ u(x,t) &= \tanh(x_0) \\ u(x,t) &= \tanh(x - u^2 t) \end{aligned}$$

(b) Determine the (explicit) solution to the linear, in-homogeneous PDE problem

$$\begin{cases} u_t + 2tu_x = 2t\cos(x), & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \operatorname{sech}^2(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{aligned} \frac{d}{dt}u(x(t),t) &= u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x &= u_t + 2tu_x \\ \frac{dx}{dt} &= 2t \end{aligned}$$

Solve the ODE

$$\frac{dx}{dt} = 2t$$
$$dx = 2tdt$$
$$\int dx = \int 2tdt$$
$$x = t^2 + x_0$$

$$\frac{du}{dt} = 2t\cos(x(t))$$
$$\int du = \int 2t\cos(t^2 + x_0)dt$$
$$u(t) = \sin(t^2 + x_0) + C$$

Initial Conditions

$$t = 0,\ u(x,0) = \operatorname{sech}^2(x)$$
$$t = 0,\ x(0) = x_0$$
$$t = 0,\ u(0) = \sin(x_0) + C$$
$$\sin(x_0) + C = \operatorname{sech}^2(x_0)$$
$$C = \operatorname{sech}^2(x_0) - \sin(x_0)$$

$$u(x_0, t) = \sin(t^2 + x_0) + \operatorname{sech}^2(x_0) - \sin(x_0)$$

$$x = t^2 + x_0$$
$$x_0 = x - t^2$$
$$u(x,t) = \sin(t^2 + (x - t^2)) + \operatorname{sech}^2(x - t^2) - \sin(x - t^2)$$

(c) Determine the (explicit) solution to the non-linear, in-homogeneous PDE problem

$$\begin{cases} u_t + uu_x = \cos(t), & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = x, & x \in \mathbb{R}. \end{cases}$$

$$\frac{d}{dt}u(x(t),t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + \frac{dx}{dt}u_x = u_t + uu_x$$
$$\frac{dx}{dt} = u$$

Solve for u

$$\frac{du}{dt} = \cos(t)$$
$$\int du = \int \cos(t) dt$$
$$u(t) = \sin(t) + C$$

Initial Conditions:

$$t = 0,\ u(0) = \sin(0) + C = C$$
$$C = x_0$$
$$u(t) = \sin(t) + x_0$$

Solve for x

$$\frac{dx}{dt} = u = \sin(t) + x_0$$
$$\int dx = \int (\sin(t) + x_0) dt$$
$$x(t) = -\cos(t) + x_0 t + C_2$$
$$t = 0,\ x(0) = -\cos(0) + x_0(0) + C_2 = -1 + C_2 = x_0$$
$$C_2 = x_0 + 1$$
$$x(t) = -\cos(t) + x_0 t + x_0 + 1$$
$$x_0(t+1) = x + \cos(t) - 1$$
$$x_0 = \frac{x + \cos(t) - 1}{t+1}$$
$$u(x,t) = \sin(t) + \frac{x + \cos(t) - 1}{t+1}$$

Question 3: Breaking Time

The breaking time t_b is the first (non-negative) time at which characteristics intersect, and we computed

$$t_b = \min_{x_0 \in \mathbb{R}} \left\{ \frac{-1}{c'(u_0(x_0))u_0'(x_0)} \geq 0 \right\} \quad \text{for} \quad \begin{cases} u_t + c(u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

For each of the following problems, determine whether the solutions breaks and if so, compute the breaking point (x_b, t_b).

(a) The non-viscous traffic model $u_t + v_1(1 - \frac{u}{u_1})u_x = 0$ with initial profile $u_0(x) = \frac{u_1}{2}e^{-(x/\beta)^2}$. How does β impact the initial profile and the breaking time?

$$c(u) = v_1\left(1 - \frac{u}{u_1}\right)$$

$$c'(u) = -\frac{v_1}{u_1}$$

$$u_0'(x_0) = \frac{u_1}{2}e^{-(x_0/\beta)^2}\left(\frac{-2x_0}{\beta^2}\right) = -u_1\frac{x_0}{\beta^2}e^{-(x_0/\beta)^2}$$

$$t_b = \frac{-1}{\left(-\frac{v_1}{u_1}\right)\left(-u_1\frac{x_0}{\beta^2}e^{-(x_0/\beta)^2}\right)}$$

We take the derivative of the denominator with respect to x_0 and set it to 0 to find the minimum time

$$\frac{d}{dx_0}\left(-\frac{v_1}{\beta^2}x_0e^{-(x_0/\beta)^2}\right) = \frac{v_1}{\beta^2}e^{-(x_0/\beta)^2}\left[1 - \frac{2x_0^2}{\beta^2}\right] = 0$$

$$1 - \frac{2x_0^2}{\beta^2} = 0 \implies x_0 = \pm\frac{\beta}{\sqrt{2}}$$

Using $x_0 = -\frac{\beta}{\sqrt{2}}$ to make the denominator positive (and t_b positive):

$$t_b = \frac{-1}{-\frac{v_1}{\beta^2}\left(\frac{\beta}{\sqrt{2}}\right)e^{-1/2}} = \frac{\beta\sqrt{2e}}{v_1}$$

$$x(t) = c(u_0)t + x_0$$

$$u_0\left(-\frac{\beta}{\sqrt{2}}\right) = \frac{u_1}{2}e^{-1/2} = \frac{u_1}{2\sqrt{e}}$$

$$c(u_0) = v_1\left(1 - \frac{1}{2\sqrt{e}}\right)$$

$$x_b = v_1\left(1 - \frac{1}{2\sqrt{e}}\right)\left(\frac{\beta\sqrt{2e}}{v_1}\right) - \frac{\beta}{\sqrt{2}}$$

$$x_b = \beta\sqrt{2e} - \frac{\beta\sqrt{2}}{2} - \frac{\beta\sqrt{2}}{2} = \beta\sqrt{2}(\sqrt{e} - 1)$$

If β is small, the change is steep, so the cars that end up stopping move fast. If β is large, the cars stop more gradually.

(b) The wacky PDE $\quad u_t + \sinh\left(\sqrt{1+\exp(u)}\right)u_x = 0$ with initial profile $u_0(x) = \tanh(x)$.

$$c(u) = \sinh\left(\sqrt{1+\exp(u)}\right)$$
$$c'(u) = \cosh\left(\sqrt{1+\exp(u)}\right)\frac{e^u}{2\sqrt{1+e^u}}$$
$$u_0(x_0) = \tanh(x_0)$$
$$u_0'(x_0) = \operatorname{sech}^2(x_0)$$

The solution breaks when $t_b = \frac{-1}{c'(u_0(x_0))u_0'(x_0)} > 0$. Because $c'(u)$ is always positive and $u_0'(x_0) = \operatorname{sech}^2(x_0)$ is always positive, the product $c'(u_0(x_0))u_0'(x_0)$ is always positive. Therefore, t_b would be negative, meaning the solution does not break for $t \geq 0$.

(c) Find a PDE $u_t+c(x,t)u_x = 0$ and initial profile $u_0(x)$ so that the solution becomes arbitrarily steep as $t \to +\infty$ but never breaks.

We want our characteristic curves to get close but never break. Let $c(x,t) = -x$.

$$u_t - xu_x = 0$$

Let $u_0(x) = \cos(x)$.

$$\frac{dx}{dt} = -x \implies \int \frac{1}{x}dx = \int -1dt$$
$$\ln|x| = -t + C \implies x(t) = x_0e^{-t}$$
$$x_0 = xe^t$$
$$u(x,t) = u_0(x_0) = \cos(xe^t)$$
$$u_x(x,t) = -\sin(xe^t)e^t$$

As $t \to \infty$, $e^t \to \infty$, making the slope u_x steep, but it never breaks.

Question 4: Understanding Qualitative Behavior

In this problem, you will (rigorously) establish several results concerning the qualitative behavior of solutions to large classes of conservation laws. Consider the PDE problem given by

$$\begin{cases} u_t + c(x,t,u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

You may assume that $c(x,t,u)$ and $u_0(x)$ are smooth functions, i.e., you can safely differentiate them.

(a) Prove that if $c = c(t)$, then the profile never deforms, i.e., $u(x,t)$ is a shift of the initial condition.

Assume $c = c(t)$. The PDE is $u_t + c(t)u_x = 0$ with $u(x,0) = u_0(x)$. Solve using the method of characteristics:

$$\frac{dx}{dt} = c(t)$$
$$dx = c(t)dt \implies x = \int c(t)dt + x_0$$
$$x_0 = x - \int c(t)dt$$
$$u(x,t) = u(x_0,0) = u_0(x_0) = u_0\left(x - \int c(t)dt\right)$$

This shows that $u(x,t)$ is just a shift of the original initial condition $u_0(x)$ by $\int c(t)dt$. The profile never deforms.

(b) Prove that if $c = c(x) > 0$, then the solution never breaks, regardless of $u_0(x)$.

Assume $c = c(x) > 0$. The characteristics are given by:

$$\frac{dx}{dt} = c(x)$$
$$\frac{du}{dt} = 0 \implies u(x,t) = u_0(x_0)$$

Since $c(x)$ is a smooth function, we can use the etxistence and uniqueness theorem for ODEs. The theorem says that for any point (x,t), there is exactly one solution curve passing through it. Therefore, characteristic curves starting at different points x_1 and x_2 can never intersect, meaning the solution never breaks.

(c) Prove that if $c = c(u) \neq$ const., then there exists an initial profile $u_0(x)$ corresponding to a solution which breaks in finite time.

Assume $c = c(u)$ is not constant. By the method of characteristics:

$$x(t) = c(u_0(x_0))t + x_0$$
$$u(x,t) = u_0(x_0)$$

Taking the partial derivative with respect to x:

$$
\begin{aligned}
u_x &= u_0'(x_0)\frac{\partial x_0}{\partial x} \\
1 &= c'(u_0(x_0))u_0'(x_0)\frac{\partial x_0}{\partial x}t + \frac{\partial x_0}{\partial x} \\
\frac{\partial x_0}{\partial x} &= \frac{1}{1 + c'(u_0(x_0))u_0'(x_0)t}
\end{aligned}
$$

The solution breaks when the denominator is zero. Thus, we need $c'(u_0(x_0))u_0'(x_0) < 0$. Because c is smooth and non-constant, there must be some u where $c'(u) \neq 0$. If $c'(u) > 0$, we can choose an initial condition $u_0(x)$ with a negative slope at that point. If $c'(u) < 0$, we can choose an initial condition $u_0(x)$ with a positive slope at that point. In either case, we can make the product $c'(u_0(x_0))u_0'(x_0)$ negative, guaranteeing that the solution will break.

(d) Prove that if $c = c(u)$ and $u_0(x)$ are both increasing functions, then the solution will never break.

Assume $c = c(u)$ and $u_0(x)$ are both increasing functions. This implies that $c'(u) > 0$ and $u_0'(x) > 0$ for all valid inputs. The breaking time is given by:

$$
t_b = \frac{-1}{c'(u_0(x_0))u_0'(x_0)}
$$

Since both c' and u_0' are positive, their product $c'(u_0(x_0))u_0'(x_0)$ is also positive. Therefore t_b will be negative. Thus, the solution will never break.